



Chapter 2

Oracle Architecture

CT208H – Oracle Database Management System



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4. Oracle Database Physical Architecture
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1. ORACLE DATABASE ARCHITECTURE

Chapter 2. Oracle Architecture



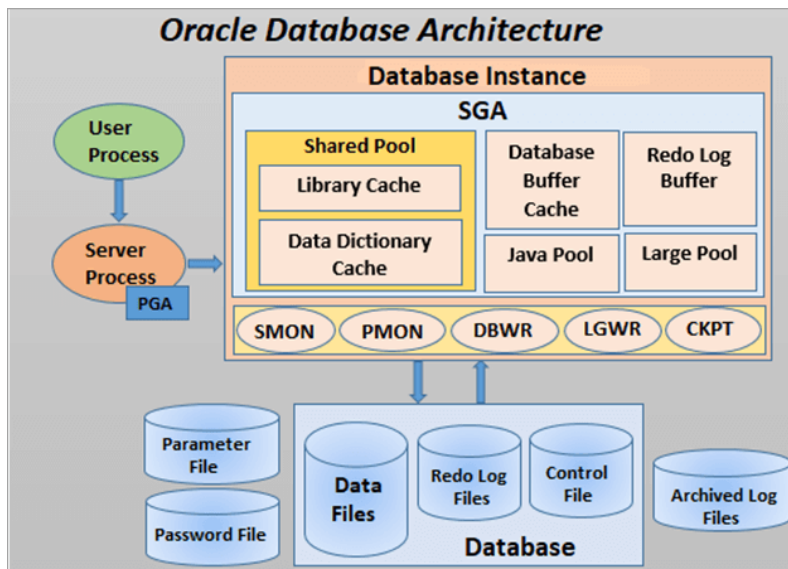
Oracle Database Architecture

- Oracle Database employs a client-server architecture:
 - The database server and client run in separate processes
 - Server process: manages data and resources
 - Client process: communicates with the server to access and manipulate data
- SQL, PL/SQL, and Java are among the programming interfaces supported by Oracle Database
- It also includes tools for DB management and administration, such as Oracle Enterprise Manager and SQL Plus



Oracle Database Architecture

- The **Oracle Database architecture** consists of:
 - Database instances (memory structure)
 - Database system
 - Processes



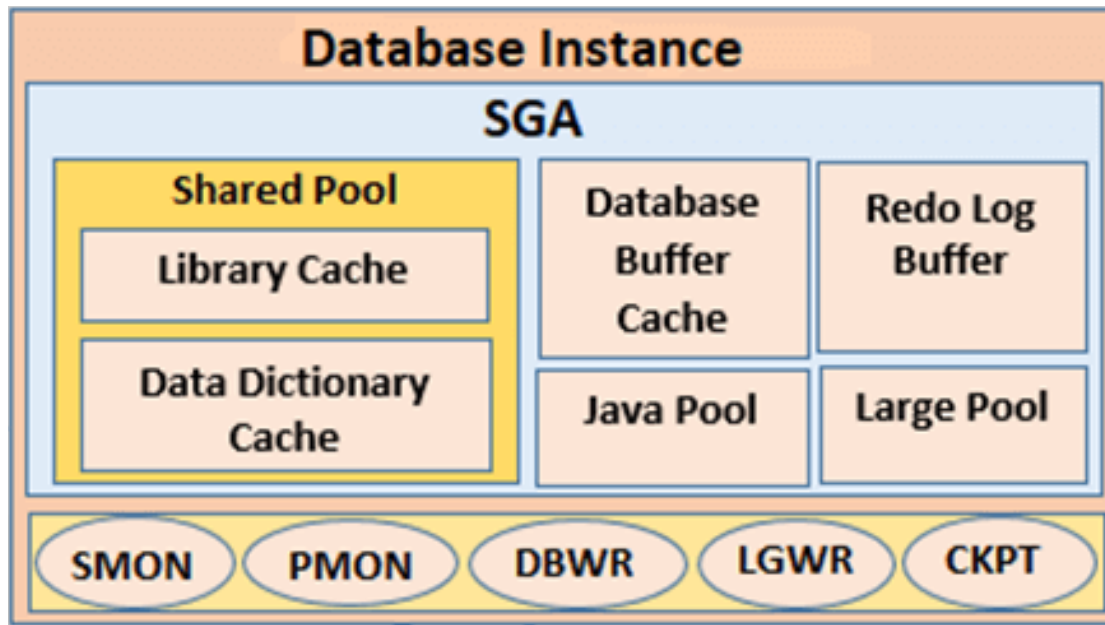


Oracle Instances

- The **instance** is a collection of **two things**:
 - System Global Area (SGA)
 - Background process
- It can be **started** and **stopped**
- When the running instance is **shut down**, all trace of its existence **goes away** at the same time
- The processes that make up the instance are known as **background processes**



Oracle Instances





System Global Area (SGA)

- A **shared memory area** of an Oracle instance
- When a **database instance starts**, SGA memory is **allocated**
- SGA stores **data** and **control information** about one database instance through its various **subcomponents**
- SGA contains three **mandatory** data structures:
 - The DB buffer cache
 - The log buffer
 - The shared pool



System Global Area (SGA)

- **Database Buffer cache:**
 - A portion of the SGA that stores **copies of data blocks** read from datafiles
 - It is used to **cache frequently accessed data blocks**, reducing the number of disks I/O required to access the data
 - This can **improve performance** by **reducing disk I/O** time and **increasing the speed** of data retrieval
 - The size of the database buffer cache **can be adjusted dynamically** and can be **automatically managed**



System Global Area (SGA)

- Redo log buffer:
 - The most crucial structure for recovery operations
 - The database's guarantee that data will never be lost
 - Consists of two or more pre-allocated files that store all changes made to the database as they occur
 - Every instance of an Oracle Database has an associated redo log to protect the database in case of an instance failure



System Global Area (SGA)

- **Shared pool:**
 - A portion of the SGA that contains **shared memory structures**, such as shared SQL and PL/SQL areas
 - It is used to store the **parsed representation** of SQL statements, **execution plans**, and **PL/SQL program** units
 - This allows for the efficient reuse of **frequently executed statements**, reducing the need for reparsing and **improving performance**



System Global Area (SGA)

- Some other optional portions of SGA:
 - Large pool:
 - It can be used for large memory allocation, such as backup and restore operations and I/O server processes
 - It is typically used to improve the performance of these operations by reducing the amount of disk I/O required
 - Java pool: used by Java Virtual Machine and related components to run Java applications



Background Process

- These processes are responsible for **managing memory**, performing **I/O operations**, and other **maintenance activities**
 - System Monitor Process (SMON): performs **system-level recovery** and **maintenance activities**
 - Process Monitor Process (PMON): monitors **other background processes**
 - Database Writer Process (DBWR): manages **data blocks**
 - Log Writer Process (LGWR): manages the **Redo blocks**
 - CheckPoint (CKPT): maintains **data files** and **control files**



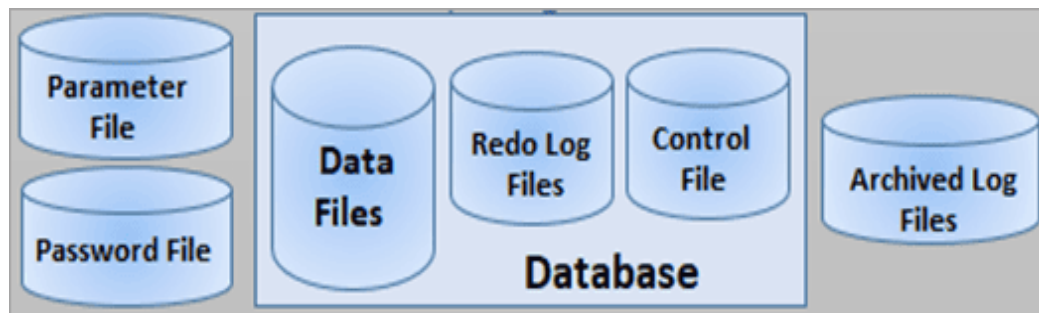
Database System

- The Database system is suited to the storage system of a computer, and it is simply the **storage of files**
- **Three categories** of files are situated in the database system:
 - Data files: hold the **actual data** in the database
 - Redo log files: hold **the changes** made in the database, used by the **recovery process** to retrieve the original information
 - Control files: hold **database status-related information** like database name, data file, and redo log file names, file locations, and log sequence number



Database System

- Other categories of files that contribute to DB management:
 - **Parameter file**: contains the **parameters** that define the way the database is expected to **start up**
 - **Password file**: holds the **user passwords** to maintain the **security** of databases





Processes

- There are **two types** of processes:
 - **User process**:
 - Also known as the **client process** or **background process**
 - Helps the users **connect to the DB instance**
 - It is stabilized when the user sends a connection request to the Oracle server
 - **Server process** (foreground process): connects the user to the database and **performs the activities on the client's behalf**, such as executing SQL statements or retrieving data from the DB



Processes

- User processes generate SQL; server processes execute SQL
- Each **server process** is associated with an area of **non-sharable memory**, called the **Program Global Area (PGA)**
 - This is a **private section** available only for server processes (while the SGA is available to all the foreground and background processes)
- **Background processes** also have PGA
- User processes usually connect to background processes through a LAN using the **Oracle Net protocol**



2. ORACLE PRODUCT STACK

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Oracle Server Family

- Three **primary groupings** of products within the server technology family:
 - **Database** server
 - **Application** server
 - **Enterprise manager** suites
- These form the basic components for Oracle's vision of **grid computing**

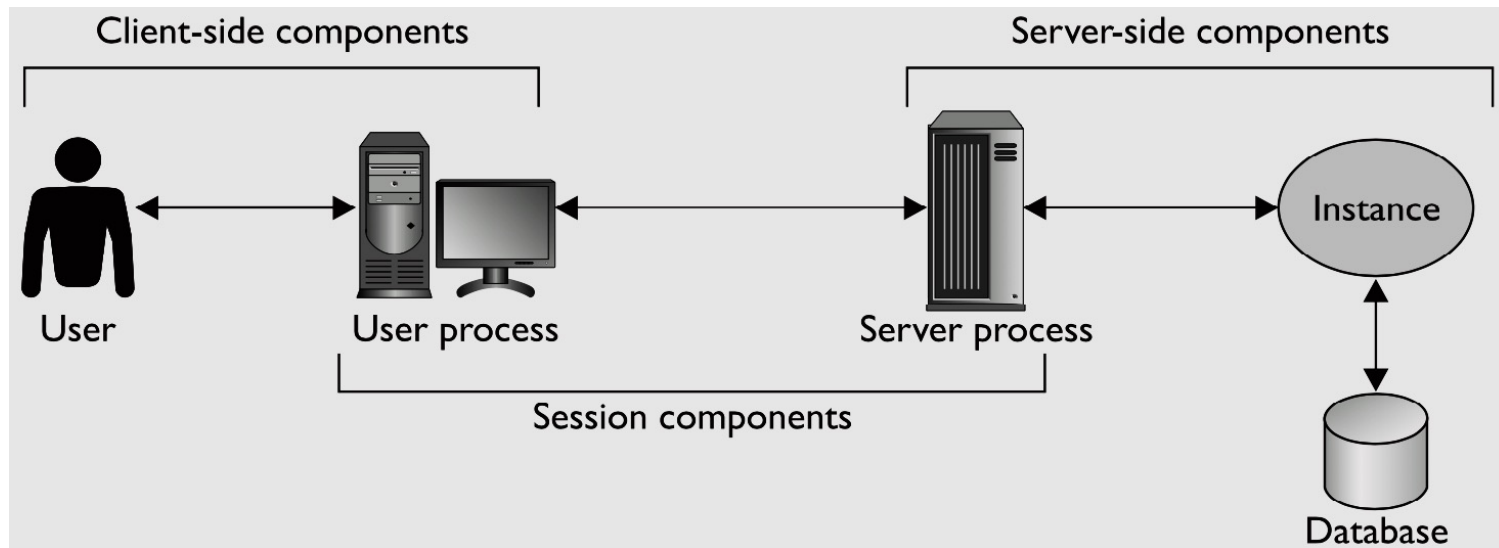


Database Server

- Comprises **Oracle instances** and **databases** with many features like Streams, Partitioning, Warehousing, Replication, and RAC
- Processing model: **client/server** (2-tier):
 - The **client tier** consists of two components: the **users** and the **user processes**
 - The **server tier** has three components: the **server processes** that execute the SQL, the **instance**, and the **database** itself



Database Server



The indirect connection between a user and a database



Database Server

- An Oracle database is a **set of files** on disk
- **Access** to the database is through the **Oracle instance**
- The instance is a set of **processes** and **memory structures**
 - It exists on the **CPU(s)** and in the **memory** of the **server node**
 - Its existence is **temporary**
- **Users** of the database **establish sessions** against the instance, which then **manages all access** to the database



Application Server

- Oracle **Application Server** provides a **platform** for developing, deploying, and managing **web applications**
- An application server allows client-side software, traditionally installed on **end-user computers**, to be replaced by applications hosted and executing from a **centralized location**
- The application **user interface** is commonly exposed to users via their **web browsers**
- These applications may use **one or more database servers**

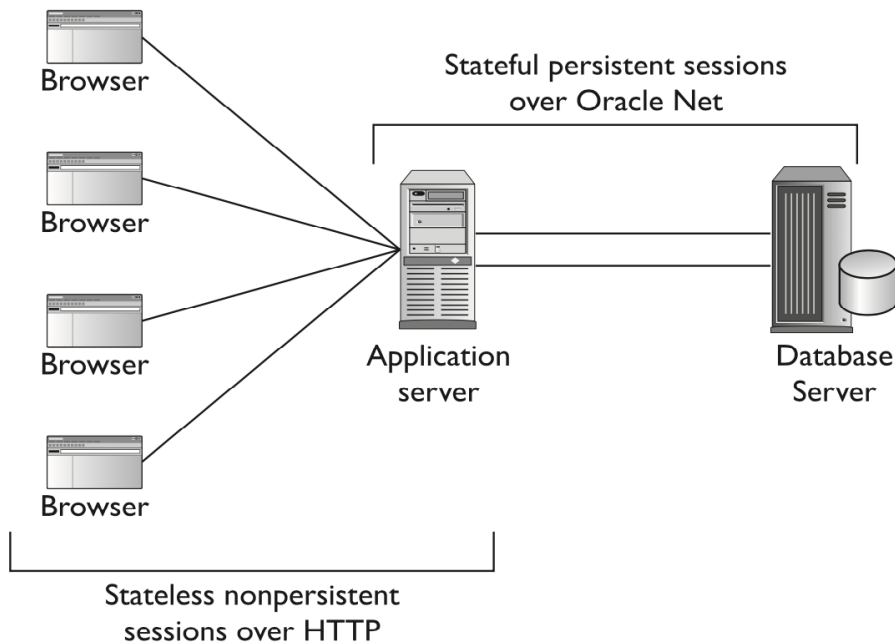


Application Server

- Web applications usually run in at least three tiers:
 - A database tier manages access to the data,
 - The client tier (often implemented via a web browser) handles the local window management for communications with the users
 - An application tier in the middle executes the program logic that generates the user interface and the SQL calls to the database



Application Server



The connection pooling model



Enterprise Manager

- Oracle **Enterprise Manager** comes in **three forms**:
 - **Database Control**: a graphical tool for managing **databases**
 - **Application Server Control**: a graphical tool for managing one or more application **server instances**
 - **Grid Control**: control globalizes the management **environment**. It enables administrators to manage hundreds/thousands of targets
- Database Control and Application Server Control consist of a **Java process** running on the server machine, which listens for **HTTP** or **HTTPS** connection requests



Oracle Development Tools

- Oracle provides several **tools** for:
 - **Developing applications**: Oracle Developer Suite (Forms, Reports, and Discoverer), Oracle Application Server,
 - **Utility programs**: Oracle E-Business Suite, Oracle Collaboration Suite,...
 - **Languages**: SQL, PL/SQL, Java. Oracle also provides the OCI (Oracle Call Interface) libraries that let code written in these languages connect to an Oracle DB and invoke SQL commands
-



3. ORACLE DATABASE LOGICAL ARCHITECTURE

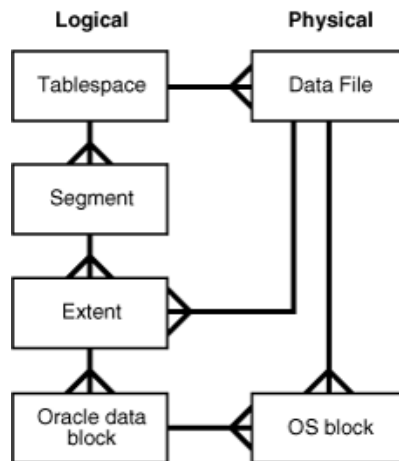
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Logical storage structure

- Oracle Database allocates **logical space** for all data in the database

- Logical units:
 - Data blocks
 - Extents
 - Segments
 - Tablespaces



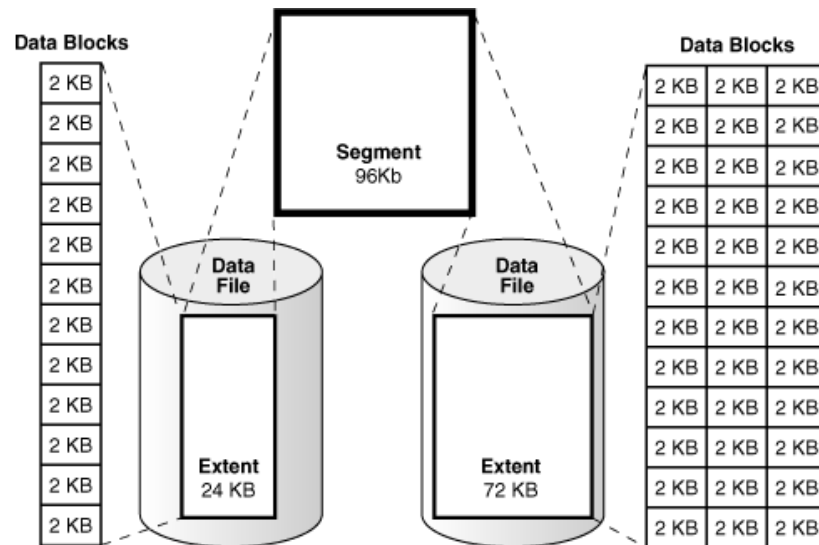
Logical and Physical Storage

- The data in the **data files** is stored in **operating system blocks**



Logical Storage Hierarchy

- A **segment** contains one or more **extents**, each of which contains multiple **data blocks**
- The figure shows the relationships among data blocks, extents, and segments within a tablespace
 - The **segment** has **two extents** stored in different data files





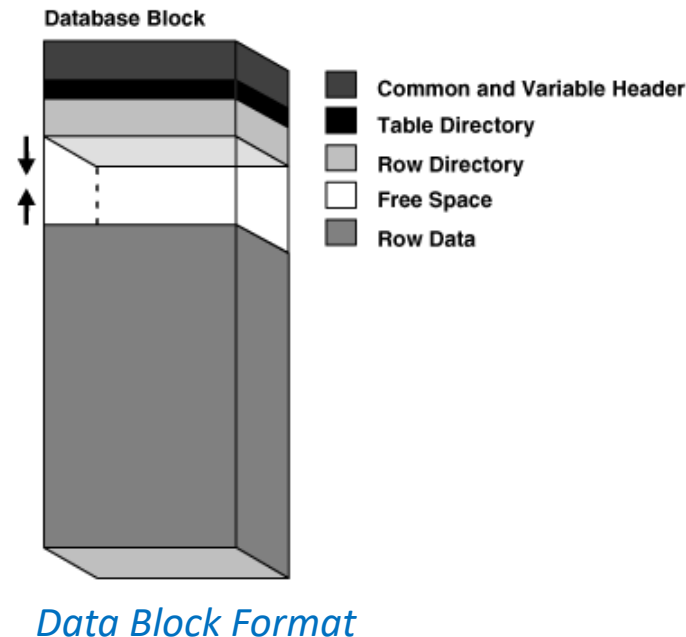
Data blocks

- A data block (or Oracle block/Page) is the **minimum unit** of database I/O
- At the **physical level**, database data is stored in **disk files** made up of **operating system blocks**
- Oracle block is a **logical storage structure** whose size and structure are **not known** to the operating system
- The **standard data block size** for a database is 4KB or 8KB (it can be set and must be a multiple of the operating system block size)



Data blocks

- Every data block has a format or **internal structure** that enables the database to track the data and free space in the block
- Oracle can use **table compression** to eliminate duplicate values in a data block



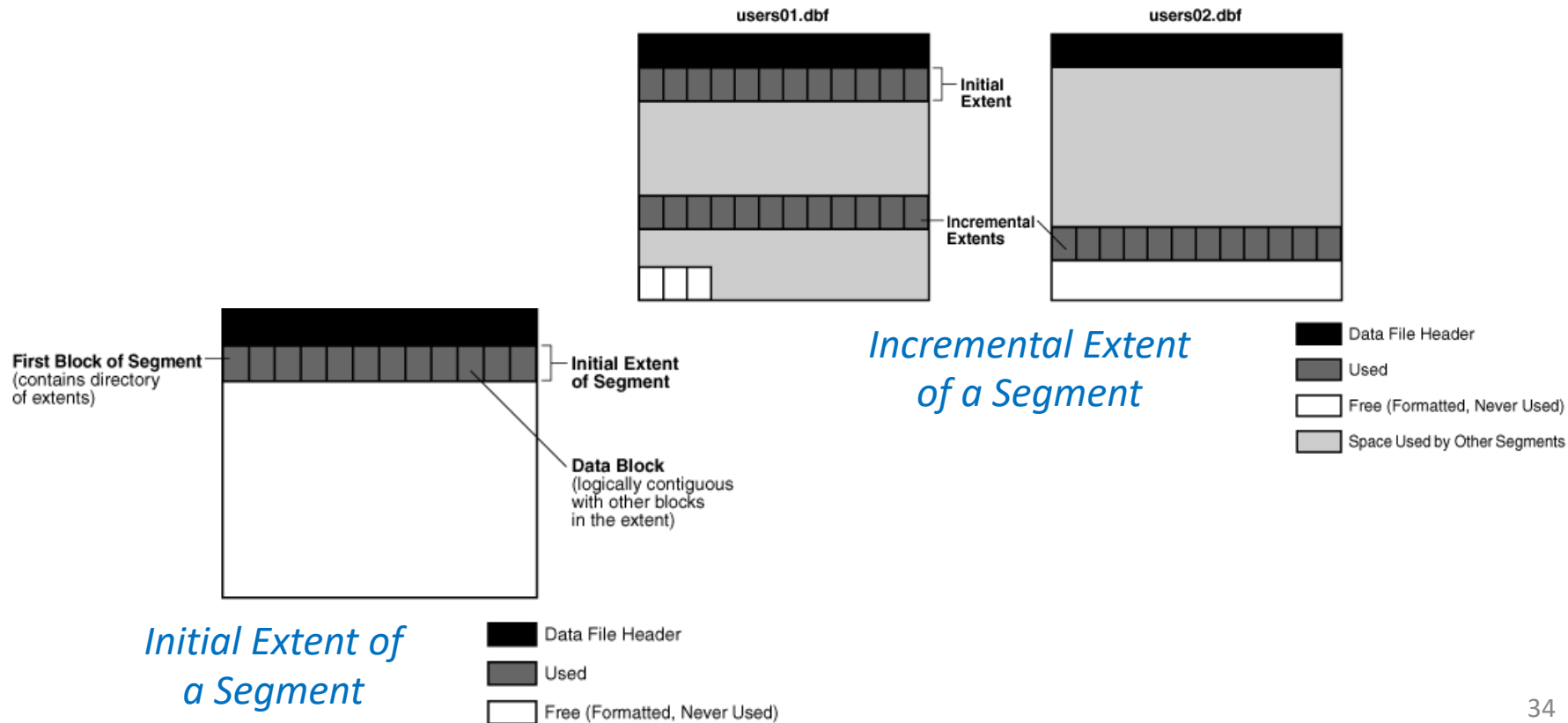


Extents

- An extent is a unit of database storage made up of logically contiguous data blocks
- Data blocks can be physically spread out on disks because of RAID striping and file system implementations
- By default, the database allocates an initial extent for a data segment when the segment is created
- An extent is always contained in one data file



Extents





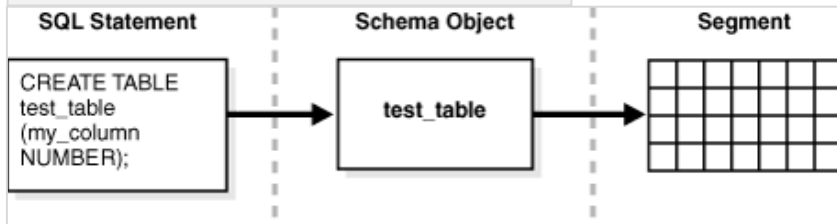
Segments

- A segment is a **set of extents** that **contains all the data** for a **logical storage structure** within a tablespace
- For example, Oracle Database allocates one or more extents to form the data segment for a table
- The database also allocates one or **more extents** to form the **index segment** for an index on a table



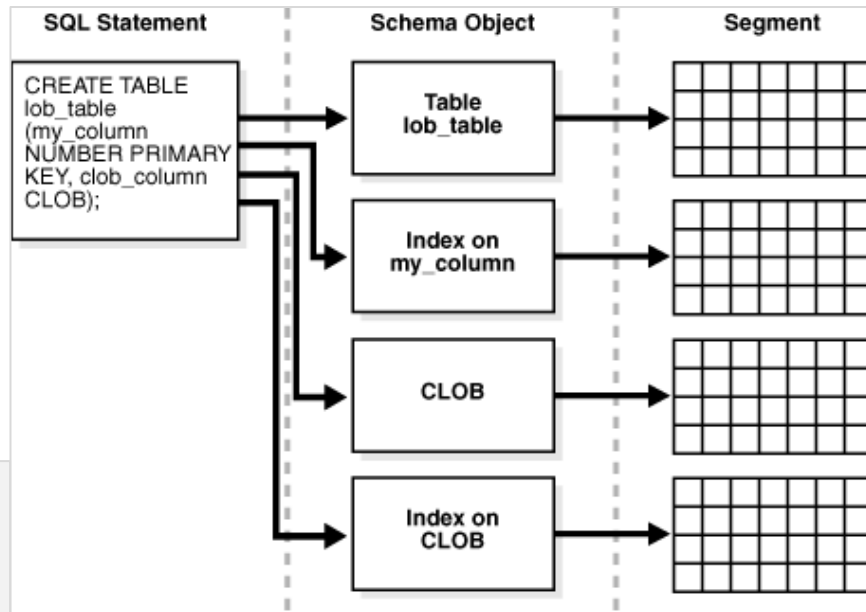
Segments

```
CREATE TABLE test_table  
(my_column NUMBER);
```



Creation of a User Segment

```
CREATE TABLE lob_table  
(my_column NUMBER PRIMARY KEY,  
 clob_column CLOB);
```



Multiple Segments



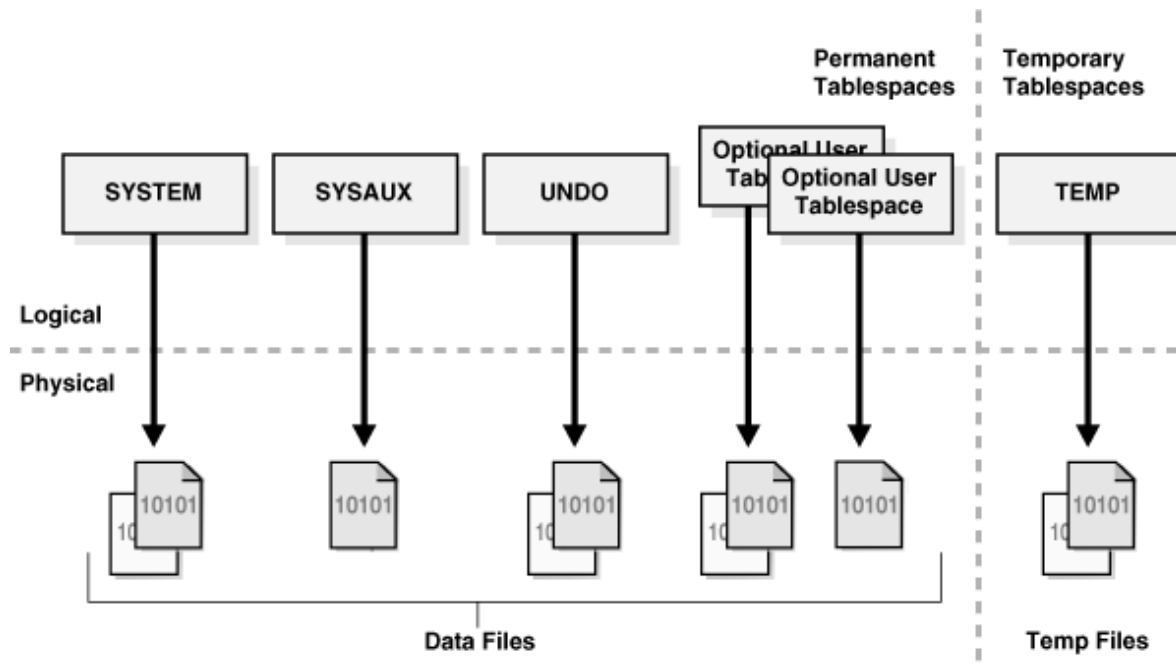
Tablespaces

- A tablespace is a **logical storage container** for segments
 - Segments are **database objects**, such as tables and indexes, that consume storage space
 - At the **physical level**, a tablespace **stores data** in one or more **data files** or temp files
- A database must have the **SYSTEM** and **SYSAUX** tablespaces
- Each **database user** is assigned a default permanent tablespace



Tablespaces

- The tablespaces in a typical database





Tablespaces

- Permanent tablespaces:
 - Groups persistent schema objects
 - The segments for objects in the tablespace are stored physically in data files
 - Each database user is assigned a default permanent tablespace
 - It is recommended that we should create at least one tablespace to store user and application data



Tablespaces

- The SYSTEM tablespace:
 - An **administrative tablespace**, included with the database
 - The SYSTEM tablespace includes the following **information**, all owned by the **SYS user**:
 - The **data dictionary**
 - **Tables and views** that contain administrative information about the database
 - **Compiled stored objects** such as triggers, procedures
 - It cannot be **renamed** or **dropped**



Tablespaces

- The SYSAUX tablespace:
 - An **auxiliary** tablespace to the **SYSTEM** tablespace
 - Helps to **reduce the load** and **enhance the performance** of the SYSTEM tablespace
 - Contains **many important Oracle system** components such as Oracle Enterprise Manager, Oracle Streams,...
 - Helps to **increase security** and **control access** to important system data by **separating** management components into SYSAUX



ORACLE®
DB Management



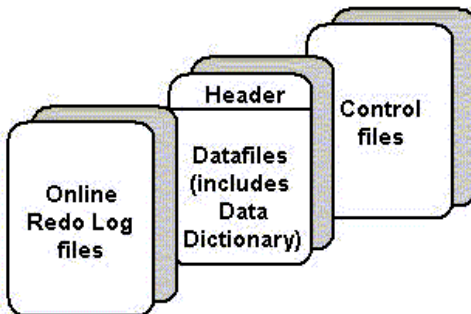
4. ORACLE DATABASE PHYSICAL ARCHITECTURE

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Database Files

- An Oracle instance consists of several physical files:
 - **Datafiles**: contain the actual data of the databases
 - **Redo log files**: contain a chronological record of changes made to the database and enable recovery when failures occur
 - **Control files**: used to synchronize all database activities





Other Key Files

- **Parameter file:** there are two types
 - The `init.ora` file (or PFILE): a **static parameter file** that contains parameters that specify how to **start up a database instance**
 - The `spfile.ora`: a **dynamic parameter file** that stores parameters to specify how to **start a database** (its parameters can be modified while the database is running)
- **Password file:** which users are **authenticated to startup/shutdown** an **Oracle instance**
- **Archived redo files:** copies of the redo log files



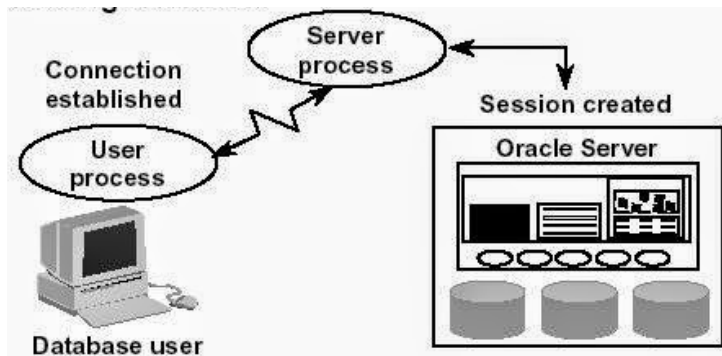
5. CONNECT TO ORACLE DATABASE

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Connect to Oracle Database

- To connect to Oracle Database, we need a **client program**, such as **SQL*Plus** or **SQL Developer**
- When connected, we can **view schema objects**, the **properties** and **data** of Oracle Database tables, and use **queries** to retrieve data from Oracle Database tables





Connect to Oracle from SQL*Plus

- To connect to Oracle Database from SQL*Plus:

1. Open the Windows **command prompt**.
2. Type **sqlplus** and then press Enter
3. Type your **user name**
4. Type your **password**

At the **SQL> prompt**, you can enter and run SQL*Plus commands, SQL statements, PL/SQL statements, and operating system commands

```
Oracle SQL*Plus
File Edit Search Options Help

SQL*Plus: Release 8.0.4.0.0 - Production on Wed Nov 19 14:56:33 1997

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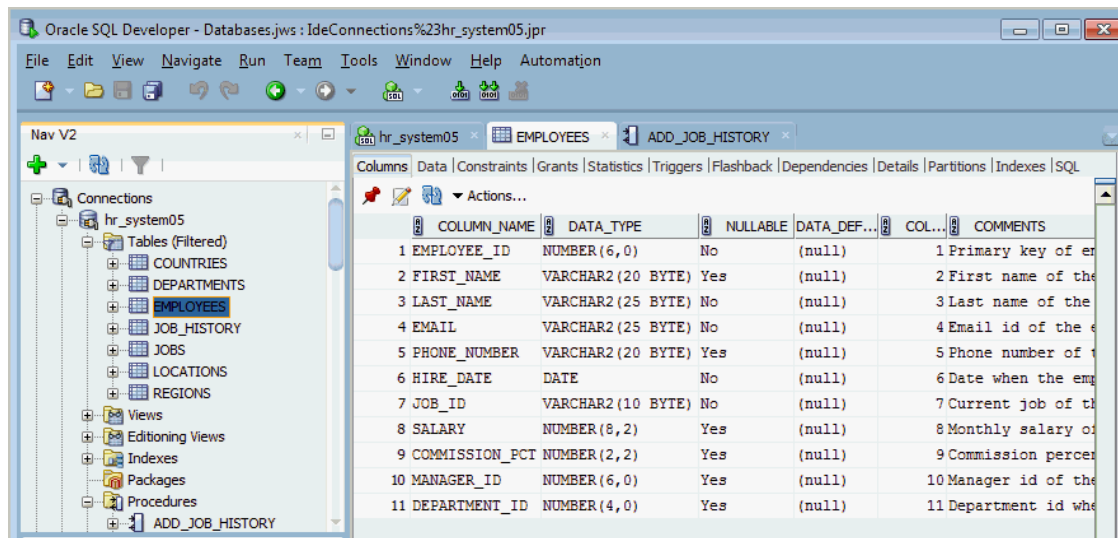
Connected to:
Oracle8 Enterprise Edition Release 8.0.4.0.0 - Production
With the Partitioning and Objects options
PL/SQL Release 8.0.4.0.0 - Production

SQL>
```




Connect to Oracle from SQL Developer

- Oracle SQL Developer is a **graphical version** of SQL*Plus that gives database developers a **convenient** way to perform basic tasks





Connect to Oracle from SQL Developer

- This tool is **included** in the Oracle Database installation
- **Installer:** <http://www.oracle.com/technetwork/developer-tools/sql-developer/>
 - Unzip the SQL Developer file into a folder (directory) of your choice, which will be referred to as **<sqldeveloper_install>**
 - Run **sqldeveloper.exe** in the **<sqldeveloper_install>\sql developer** folder
- Oracle SQL Developer installation **guide:**
<https://docs.oracle.com/en/database/oracle/sql-developer/18.2/rptig/installing-sql-developer.html#RPTIG-GUID-A3509B92-90A4-4268-8027-FE85DE5554A8>



6. ORACLE DATABASE SQL EXECUTION

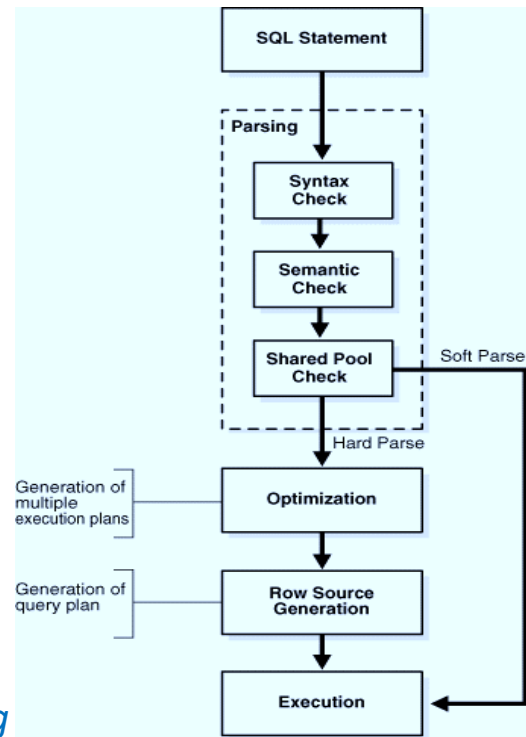
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About SQL processing

- SQL processing is the parsing, optimization, row source generation, and execution of a SQL statement
- The figure depicts the general stages of SQL processing
- Depending on the statement, the database may omit some of these stages

Stages of SQL Processing





SQL Parsing

- The **first state** of SQL processing
- Involves **separating** the pieces of a SQL statement **into a data structure** that other routines can process
- The applications make a **parse call** to the database to prepare the statement for execution
- The parse call creates a **cursor** which is a handle for the **session-specific private SQL area** that holds a parsed SQL statement and other processing information



SQL Parsing

- **Syntax check:**
 - Check the **syntactic validity**
 - A statement that **breaks a rule for well-formed SQL syntax** **fails** the check
 - For example, the following statement **fails** because the keyword FROM is **misspelled** as FORM:

```
SQL> SELECT * FORM employees;  
SELECT * FORM employees  
          *  
ERROR at line 1:  
ORA-00923: FROM keyword not found where expected
```



SQL Parsing

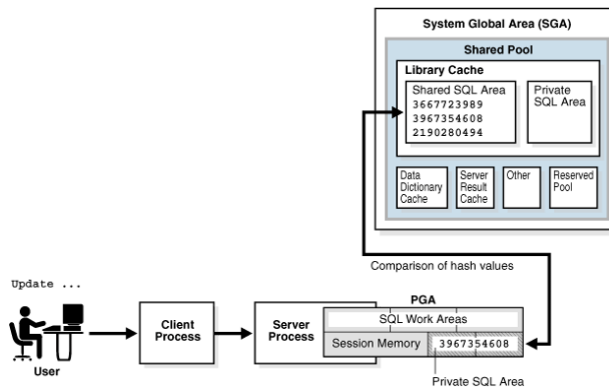
- Semantic check:
 - The semantics of a statement are its meaning
 - A semantic check determines whether a statement is meaningful
 - For example, whether the objects and columns in the statement exist

```
SQL> SELECT * FROM nonexistent_table;  
SELECT * FROM nonexistent_table  
          *  
ERROR at line 1:  
ORA-00942: table or view does not exist
```



SQL Parsing

- **Shared Pool check:**
 - To determine whether it **can skip resource-intensive steps** of statement processing
 - When a user submits an SQL statement, the database searches the shared SQL area to see if **an existing parsed statement** has the same **hash value**





SQL Optimization

- Improves the **efficiency** and **performance** of SQL queries to ensure SQL statements execute as quickly and efficiently as possible
- Just for **DML** statements
- **Exception** only occurs when the **DDL includes a DML** component, such as a **subquery** that requires optimization



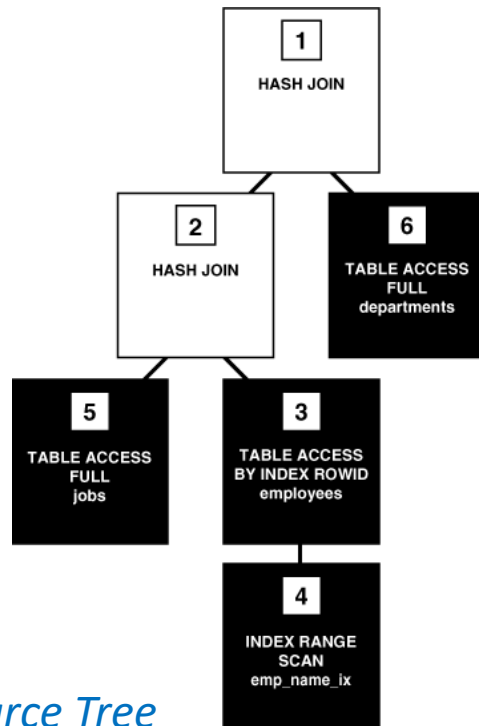
SQL Row Source Generation

- A software that receives the **optimal execution plan** from the optimizer and produces an **iterative execution plan** that is usable by the rest of the database
- Iterative execution plan:
 - The plan takes the form of a **combination of steps**
 - Each step returns a **row set**
 - A row source is a **row set** returned by a step in the execution plan, along with a **control structure** that can **iteratively process** the rows



SQL Execution

- SQL engine executes each **row source** in the **tree** produced by the **row source generator**
 - The order of the steps in execution is the **reverse** of the order in the plan
 - Each node of the tree acts as a **row source** (each step of the execution plan either retrieves rows from the database or accepts rows from one or more row sources as input)



Row Source Tree



QUESTIONS?

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